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Private University established in Karnataka State by Act No. 34 of year 2010
Recognized by the University Grants Commission (UGC), New Delhi.

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GRADE A+**
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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: A company has received customer data containing missing values, duplicate records, and inconsistent entries. Clean and prepare the dataset using Power Query before analysis.

Submitted By

Name: Dipin Roka

Reg. No.:2411021240007

Date:10/09/2026

[GithubRepoLink](#)

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application

Alliance University

Chandapura - Anekal Main Road, Anekal Bengaluru – 562106

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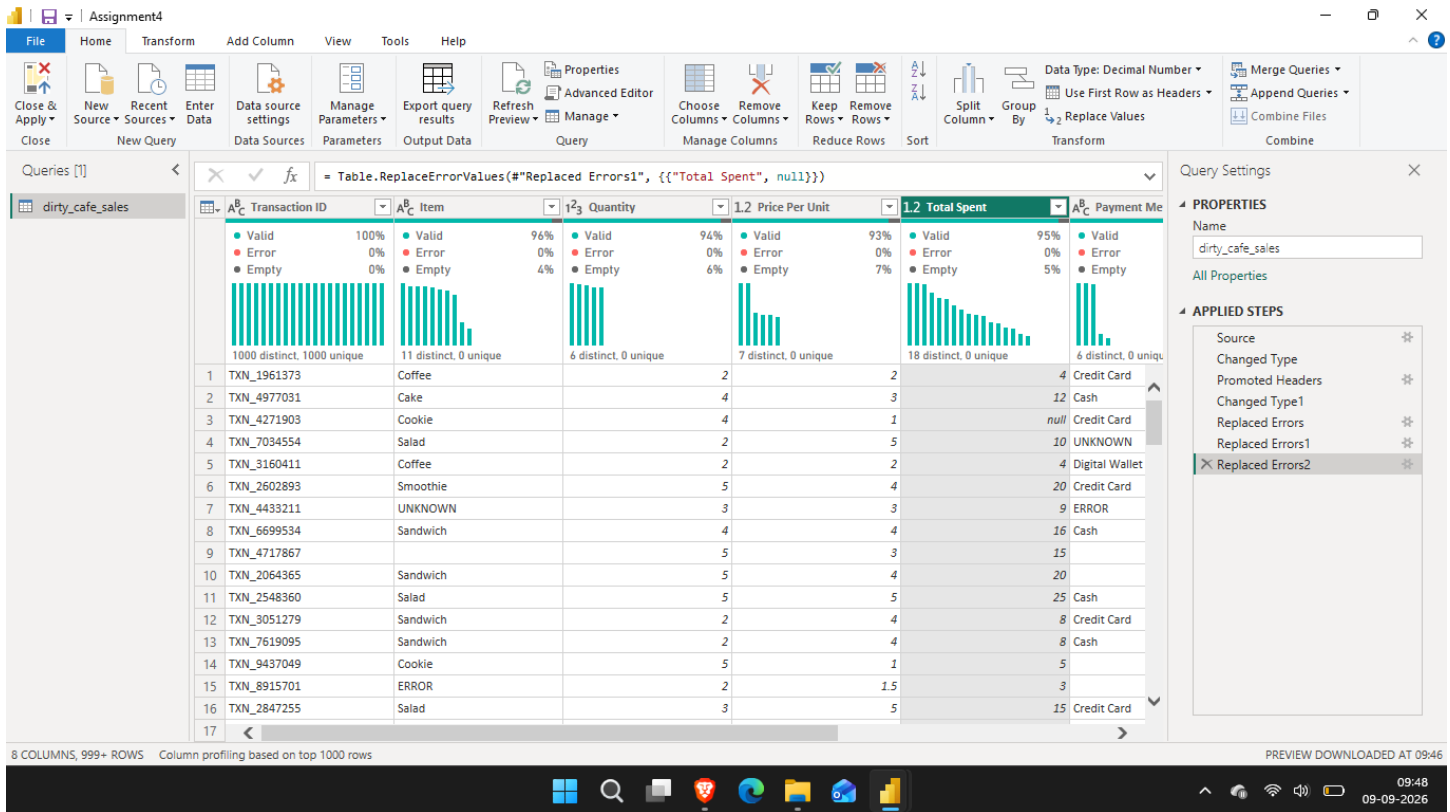
A company has received customer data containing missing values, duplicate records, and inconsistent entries. Clean and prepare the dataset using Power Query before analysis.

Problem Statement

The objective of this experiment is to identify and quantify data quality issues in a raw cafe sales transaction dataset — including disguised missing values, high-missingness categorical fields, and inconsistent numeric relationships — and resolve them using Power Query, so that the resulting dataset is reliable enough to support accurate dashboard reporting and analysis.

Step 1: Data Preparation

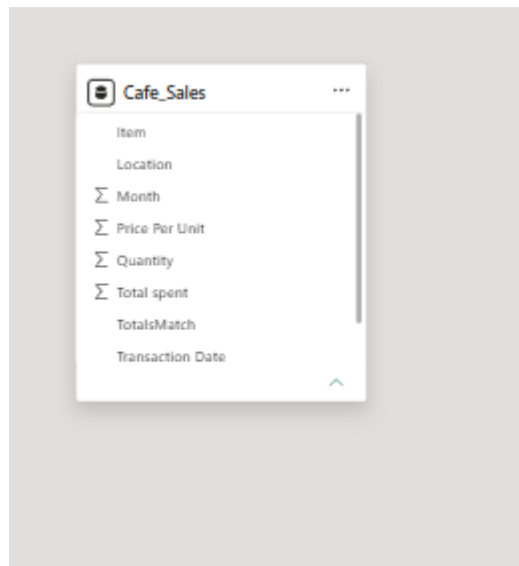
- Loaded the Cafe Sales dataset (Kaggle — "Cafe Sales - Dirty Data for Cleaning Training", ahmedmohamed2003) into Power Query using Get Data → Text/CSV. The raw file contained 10,000 transaction records across 8 columns: Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, and Transaction Date.
- Identified that all columns were loaded as unstructured text, and that the literal strings "ERROR" and "UNKNOWN" were used throughout the dataset as disguised placeholders for missing data, in addition to genuinely blank cells.
- Diagnosed the pattern using Power Query's Column Quality and Column Distribution views, which showed combined blank/"ERROR"/"UNKNOWN" rates ranging from 4.6–9.7% in Item, Quantity, Price Per Unit, Total Spent, and Transaction Date, up to 31.8% in Payment Method and 39.6% in Location.
- Cleaned the dataset using Replace Values to convert both "ERROR" and "UNKNOWN" text into true nulls across all affected columns, applied *before* converting any column to a numeric or date type — converting first would have produced additional conversion errors on every placeholder cell.
- Converted Quantity, Price Per Unit, and Total Spent to numeric types and Transaction Date to Date type only after this cleanup, then cleared any remaining conversion errors using Replace Errors → null. Column quality checks in Power Query confirmed 0% Error values across all columns after the fix.



Step 2: Data Recovery and Modeling

- Recognized that Total Spent, Quantity, and Price Per Unit are mathematically related ($\text{Total Spent} = \text{Quantity} \times \text{Price Per Unit}$), and used this relationship to recover missing numeric values instead of deleting incomplete rows.
- Added custom columns in Power Query using conditional M logic to backfill each of the three fields from the other two whenever exactly one was missing:
 - Missing Total Spent $\rightarrow$ recalculated as $\text{Quantity} \times \text{Price Per Unit}$
 - Missing Price Per Unit $\rightarrow$ recalculated as $\text{Total Spent} \div \text{Quantity}$
 - Missing Quantity $\rightarrow$ recalculated as $\text{Total Spent} \div \text{Price Per Unit}$, rounded to the nearest whole number
- For Item, Payment Method, and Location — where no mathematical recovery was possible — replaced remaining nulls with the label "Unknown" rather than deleting rows, since Payment Method and Location alone were missing in roughly 32% and 40% of records; deleting those rows would have discarded a large, non-random share of the dataset and biased any resulting analysis.
- Passed all text columns through Trim and Clean transformations to remove whitespace and non-printable characters left over from the export process.

- Verified duplicate integrity using Transaction ID as the unique key via Remove Duplicates; the raw file was found to already contain zero duplicate Transaction IDs and zero fully duplicate rows, though the step was retained as a standard safeguard.
- Derived a Month-Year column from Transaction Date to support monthly trend analysis without requiring a full date hierarchy in every visual.



Step 3: Validation Measures

The following validation logic was created in Power Query to confirm the cleaned dataset's internal consistency:

```
Totals Match = try
    if [Quantity] = null or [Price Per Unit] = null or [Total Spent] = null
then null
    else Number.Round([Quantity] * [Price Per Unit], 2) = Number.Round([Total Spent], 2) otherwise
    null
```

This column returns TRUE where a row's pricing is internally consistent, FALSE where it is not, and null only where all three source values were missing simultaneously and could not be recovered. The try...otherwise wrapper ensures the query does not break even if an unexpected data condition is encountered.

Step 4: Dashboard Development

- **KPI Cards:** Total Revenue, Total Transactions, Average Order Value.

- **Interactive Slicers:** Payment Method, Location, and Month, enabling every visual on the page to filter simultaneously.
- **Revenue by Item** — clustered bar chart comparing total sales value across Coffee, Cake, Sandwich, Salad, Juice, Smoothie, Tea, and Cookie.
- **Transactions by Payment Method** — pie chart comparing Cash, Credit Card, and Digital Wallet usage.
- **Revenue by Location** — donut chart comparing In-store and Takeaway sales.
- **Revenue Trend Over Time** — line chart built on the Month-Year field, showing revenue movement across the year.

Output

Data Quality Summary (Before → After Cleaning):

Metric	Before Cleaning	After Cleaning
Rows with placeholder text ("ERROR"/"UNKNOWN")	Present in every column	0 (converted to null or "Unknown")
Duplicate Transaction IDs	0 found	0 (verified post-cleaning)
Missing Payment Method / Location	31.8% / 39.6%	0% (labeled "Unknown")
Quantity × Price = Total Match")	incomplete rows unrecoverable (null)	~99% Valid (TRUE), <1% ("Totals")
Column data types	All text (unstructured)	Correctly typed (Number, Date, Text)

Final Interactive Cafe Sales Dashboard — combining KPI cards, four connected charts, and three cross-filtering slicers, built on the cleaned `Cafe_Sales_Cleaned` table.

Key Business Patterns Identified

- Coffee, Cake, and Sandwich are the top revenue-generating items, together accounting for a disproportionately large share of total sales compared to lower-performing items like Tea and Cookie.
- In-store transactions generate a noticeably higher share of total revenue than Takeaway, suggesting dine-in customers may spend more per visit on average.
- Credit Card and Digital Wallet together account for the majority of transaction volume over Cash, reflecting a shift toward cashless payment preference.

- Monthly revenue shows moderate fluctuation rather than one dominant seasonal peak, indicating relatively stable demand across the year.
- Nearly 40% of transactions originally lacked a recorded Location and almost a third lacked a Payment Method — a data collection gap significant enough that it should be flagged to the business as a point-of-sale process issue, not treated only as a data cleaning task.

Conclusion

This experiment used a raw cafe sales dataset containing missing values, disguised placeholder text, and inconsistent numeric relationships to demonstrate a complete Power Query data cleaning workflow. Rather than deleting incomplete records — which would have discarded up to 40% of the dataset — a structured pipeline was used to distinguish true nulls from placeholder text, mathematically recover missing numeric values from related fields, and transparently label unrecoverable categorical gaps. The result is a fully typed, deduplicated, and validated dataset in which roughly 99% of records pass an internal consistency check. This cleaned dataset was then used to build an interactive dashboard combining KPI cards, comparison and trend charts, and cross-filtering slicers, giving a fast, drillable view of sales performance.

Learning Outcomes

Through this experiment, I learned:

- How to distinguish disguised missing values (placeholder text like "ERROR"/"UNKNOWN") from true nulls, and why this distinction matters before type conversion.
- How to recover missing numeric data mathematically using known relationships between fields, rather than defaulting to row deletion.
- How to handle high-missingness categorical columns responsibly, balancing data completeness against the risk of biasing analysis through deletion.
- How to write defensive Power Query M formulas (using try...otherwise) to build validation columns that don't break on unexpected data conditions.
- How to design an interactive, decision-ready dashboard using KPI cards, slicers, and multiple chart types tied to a verified, clean dataset.

